Akash Nandi

Email: akashnandi556@gmail.com

LinkedIn

Phone N.o: +918093541513

SUMMARY

Machine Learning Engineer with expertise in NLP, supervised and unsupervised learning, and statistical analysis. Skilled in feature engineering, including handling missing values, imbalanced data, and advanced encoding techniques. Proficient in Python, SQL, and ETL/ELT processes using SSIS for data transformation and integration. Experienced in data visualization with Power BI and leveraging Google Cloud and Big Query for scalable analytics. Strong team collaborator with solid analytical thinking and Agile methodology experience, focused on developing data-driven solutions for business optimization and predictive modelling.

SKILLS

- **GEN-AI** (RAG, Agentic RAG, Langraph)
- **NLP** (TF-IDF, NLTK, LSTM, GRU RNN, Transformers)
- **Machine Learning:** Supervised and Unsupervised Machine Learning
- **Feature Engineering:** Missing value treatment, Imbalanced dataset handling, Encoding techniques (OHE, label, ordinal, target-guided).
- **Statistical Analysis & Analytics:** Descriptive, predictive, and prescriptive analytics.
- Agile methodologies, strong analytical thinking, team collaboration.
- **Programming & Querying:** Python (OOPS, functional programming), SQL(Microsoft SQL Server).
- **Data Processing:** ETL/ELT **(SSIS)**, data cleaning, data wrangling.
- **Data Visualization & Reporting:** Power BI.
- **Cloud:** Google Cloud Services, Google Big Query.

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Full Stack Data Analytics from iNeuron.ai
- MYSQL for Data Analytics and BI (Udemy)
- Microsoft Certified Azure Fundamentals
- Hacker-Rank Certifications: Python, SQL
- Microsoft Certified: Azure Data Fundamentals

EDUCATION

- Bachelor of Technology (Computer Science and Information Technology),
  Siksha 'O' Anusandhan University, Bhubaneswar | CGPA-8.35 | 2017 – 2021
- Higher Secondary
  Dalmia Vidya Mandir, Rajgangpur | 71% (PCA) | 2015 – 2017
- Secondary
  Dalmia Vidya Mandir, Rajgangpur | CGPA-9.2 | 2014 – 2015

WORK EXPERIENCE

- **Jul'21 –May'23 | Analyst | Capgemini, Pune**

  Key Result Areas:

  - Utilizing skills in Python Scripting and SQL for the design, development and deployment of new features into existing products.
  - Managing the maintenance of program modules including operational support, production support, preventative and corrective maintenance and enhancements.
  - Conducting code reviews, analysing and modifying existing codes for application optimization and writing new codes as required.

- Performing analysis, design, verification, demonstration and maintenance of products in a process-driven, team environment.
- Troubleshooting errors and resolving customer issues raised in GitLabs and JIRA.
- Ensuring delivered solutions are compatible with the user's environment and can run smoothly across multiple platforms.

➢ **July'23 –Till Now | ML Engineer | TCS, Pune**

Key Result Areas:

- Using **Python** to extract the raw data and clean it as per the requirement.
- Conduct **Exploratory Data Analysis** to identify trends, patterns and anomalies, facilitating informed decision-making purposes.
- Worked on **Subscriber Churn Prediction Model** to identify the users that were likely to cancel their Subscriptions. Classification Models like **Logistic Regression**, **Random Forest** and **XGBoost** ML Algorithms were used for the prediction of **High-Risk Users**.
- Using **SQL, GCP, Power BI** for Data Analysis, Warehousing and Visualization.
- Created an **ETL pipelines** in SSIS to automate the process of inserting CSV files in Microsoft SQL Server database.
- Collaborate with cross functional teams to identify the business needs and translate them into actionable data analysis solutions.

PROJECTS

➢ **Title: <u>Algerian Forest Fire Prediction</u>**

- **Description:** Conducted an in-depth analysis of the Algerian Forest Fire dataset to identify key environmental factors influencing fire occurrence. Performed data cleaning, exploratory data analysis (**EDA**), and visualizations using Python libraries (Pandas, Matplotlib, Seaborn). Developed predictive models using machine learning algorithms such as **Logistic Regression, Random Forest and SVM** to classify fire-prone areas. Achieved high model accuracy and provided actionable insights for fire risk mitigation.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook, VS Code.

➢ **Title: <u>Prediction of Heart Attack</u>**

- **Description:** Five million Americans are living with heat diseases and the numbers are expected to rise. It is very important to understand the factors which causes heart attacks so that precaution can be taken by the individuals. In order to understand the reason of heart attack, a data was collected from various hospitals across US.
- **Tech Stack:** Python, Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn.

➢ **Title: <u>RAG Document Search System</u>**

- **Description:** Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline to enable intelligent document search and question answering. Implemented document ingestion, chunking, and embedding generation to store vector representations in a vector database. Designed a retrieval mechanism to fetch semantically relevant chunks based on user queries, which are then passed to a Large Language Model to generate accurate, context-aware responses.
- **Tech Stack:** Python, LangChain, OpenAI API, FAISS / ChromaDB, Hugging Face Transformers, Ollama Embeddings, Vector Database, Jupyter Notebook.